

BYUNGWOO JEON

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RESEARCH INTEREST

The north star of my research is to develop a unified and scalable framework aligned with multi-modalities and human preferences for various vision-centric tasks. To this end, I am delving into multi-modal understanding (e.g., VLM, UMM) and robot learning (e.g., VLA), with a focus on 3D/4D world modeling that bridges perception, cognition and action for real-world embodied intelligence.

- **Multi-modal learning**
 - Reasoning
 - Omni-modal generation
 - World modeling
- **High-dimensional vision**
 - Spatio-temporal understanding
 - Motion & pose estimation
- **Robot learning**
 - Long-horizon planning
 - World action modeling

PUBLICATIONS

C: conference, J: journal, W: workshop, P: preprint / * equal contribution, † equal advising

- [C3] **SpatialBoost: Enhancing Visual Representation through Language-Guided Reasoning**
Byungwoo Jeon^{*}, Dongyoung Kim^{*}, Huiwon Jang, Insoo Kim, Jinwoo Shin
European Conference on Computer Vision (ECCV), 2026
- [C2] **Vision-aligned Latent Reasoning for Multi-modal Large Language Model**
Byungwoo Jeon, Yoonwoo Jeong, Hyunseok Lee, Minsu Cho[†], Jinwoo Shin[†]
International Conference on Machine Learning (ICML), 2026
- [C1] **TrackIME: Enhanced Video Point Tracking via Instance Motion Estimation**
Seong Hyeon Park, Huiwon Jang, Byungwoo Jeon, Sukmin Yun, Paul Hongsuck Seo, Jinwoo Shin
*Conference on Neural Information Processing Systems (NeurIPS), 2024, **Spotlight Presentation (326/15671=2.08%)***
- [P5] **Latent Reasoning in Continuous Space for Unified Multimodal Models**
Byungwoo Jeon, Yoonwoo Jeong, Hyunseok Lee, Minsu Cho[†], Jinwoo Shin[†]
Under review
- [P4] **Physically Plausible Multi-View Object Swapping for Robot Data Augmentation**
Jonghoon Lee^{*}, Seong Hyeon Park^{*}, Byungwoo Jeon, Minha Lee, Jinwoo Shin
Under review
- [P3] **Cog3DMap: Multi-View Vision-Language Reasoning with 3D Cognitive Maps**
Chanyoung Gwak^{*}, Yoonwoo Jeong^{*}, Byungwoo Jeon, Hyunseok Lee, Jinwoo Shin[†], Minsu Cho[†]
Under review
- [P2] **Visual Tokenization with Hierarchical Abstraction for Unified Multimodal Modeling**
JB Kim, Hyeonmo Kang, Byungwoo Jeon, June Suk Choi, Kyungmin Lee, Jinwoo Shin[†], Jeonghoon Kim[†]
Under review
- [P1] **CADS: Consistent and Accurate Depth Estimation via Synthesizing Depth Maps with Image-based Rendering**
Subin Kim^{*}, Seong Hyeon Park^{*}, Byungwoo Jeon, Sihyun Yu, Kihyuk Sohn, Jinwoo Shin
Under review
- [W1] **Hierarchical Vision Language Action Model Using Success and Failure Demonstrations**
Jeongeun Park, Jihwan Yoon, Byungwoo Jeon, Juhan Park, Jinwoo Shin, Kyungjae Lee, Namhoon Cho, Sangdoo Yun, Sungjoon Choi
CoRL Workshop on Safe and Robust Robot Learning for Operation in the Real World, 2025

EDUCATION

KAIST, Ph.D. student in Artificial Intelligence (advisor: Prof. Jinwoo Shin)	Sep 2024 - Current
Korea University, B.S. in Computer Science and Engineering, Statistics	Mar 2020 - Aug 2024

WORK EXPERIENCE

ALINLAB, Research Intern (advisor: Prof. Jinwoo Shin)	Mar 2023 - Aug 2024
Arcreal, ML Engineer (host: Prof. Jinwoo Shin)	Jan 2023 - Jan 2024
MLVLAB, Research Intern (advisor: Prof. Hyunwoo J. Kim)	Jul 2022 - Dec 2022
M-monstar, Fullstack Engineer	Jul 2021 - Aug 2021

EXTERNAL COLLABORATOR

POSTECH CVLAB, Working on VLM reasoning [C2, P5, P3] (advisor: Prof. Minsu Cho)	Apr 2025 - Current
Korea Univ. RILAB, Working on VLA reasoning [W1] (advisor: Prof. Sungjoon Choi)	Jul 2025 - Current
ETH-Zürich CVG, Working on 4D world modeling (advisor: Dr. Sunghwan Hong)	Jul 2026 - Current

PROJECTS

Industry Projects

NVIDIA, Latent World Models (2026–)
RLWRLD, RLDX: Vision-Language-Action Model for Dexterous Manipulation (2026–)
SAMSUNG, Semiconductor Pattern Generation (2025–2027)

Personal Projects

Google Developer Groups, Google Solution Challenge: Recovery (2024), digiHow (2023)
Google Developer Groups, Horang Studio: Diffusion-based personalized AI profile service (2023)

ACADEMIC SERVICES

Conference Reviewer: ECCV'26
Workshop Reviewer: Space in Vision, Language, and Embodied AI (SpaVLE) @ NeurIPS'25

HONORS & AWARDS

Silver Prize, K-Data Science Hackathon, 2023 (\$2,000)
Top 8, Artificial Intelligence Grand Challenge, 2022
Dean's List, Korea University, 2020–2023

EXTRACURRICULAR ACTIVITIES

Google Developer Groups on Campus (GDGoC) @ Korea University, Lead and Founder	Aug 2022 - Jul 2023
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SELECTED TALKS

Korea Univ., GDGoC, "Meta Runner on AI," Mar 2026
Yonsei & Korea Univ., GDGoC, "Towards the New World of Computing: CV on Real-world," Dec 2024
Korea Univ., GDGoC, "Recent Advances in Machine Learning," Nov 2023

MENTORING

- **Byungjun Yoon**, Ph.D. student at KAIST AI
Topic: Atomic VLA
Mar 2025 - Current
- **Jonghoon Lee**, M.S. student at KAIST AI
Topic: 3D/4D mesh generation [P4]
Apr 2025 - Current
- **Minha Lee**, M.S. student at KAIST AI
Topic: 3D/4D mesh generation [P4], Atomic world modeling
Mar 2026 - Current
- **Kiyoung Kwon**, now senior undergrad at SNU AI
Topic: Agentic AI4Science
Jan 2026 - Current
- **Hyunseo Yang**, now senior undergrad at SNU CSE
Topic: GUI agent with world modeling
Jul 2026 - Current
- **Heejun Jung**, now senior undergrad at KAIST SoC
Topic: Scalability of world modeling
Jul 2026 - Current
- **Seoyeon Byun**, M.S. student at KAIST AI
Topic: 3D/4D neural rendering, **Collaborate**: GDGoC 3rd Lead
Aug 2022 - Aug 2024
- **Seonghu Jeon**, M.S. student at KAIST AI
Topic: 3D/4D neural rendering, **Collaborate**: GDGoC 3rd AI Core
Mar 2022 - Aug 2024

REFERENCE

Jinwoo Shin, Full Professor at KAIST

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Sungjoon Choi, Assistant Professor at Korea Univ.

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Sunghwan Hong, Postdoc fellow at ETH-Zürich

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Seong Hyeon Park, Visiting scholar at NYU

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Yoonwoo Jeong, Ph.D. student at POSTECH

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